

TMA4315 Exercise 1 Linear models for Gaussian data

Thomas Guillot , Lucas Paraveman , Dario Mazotti

Contents

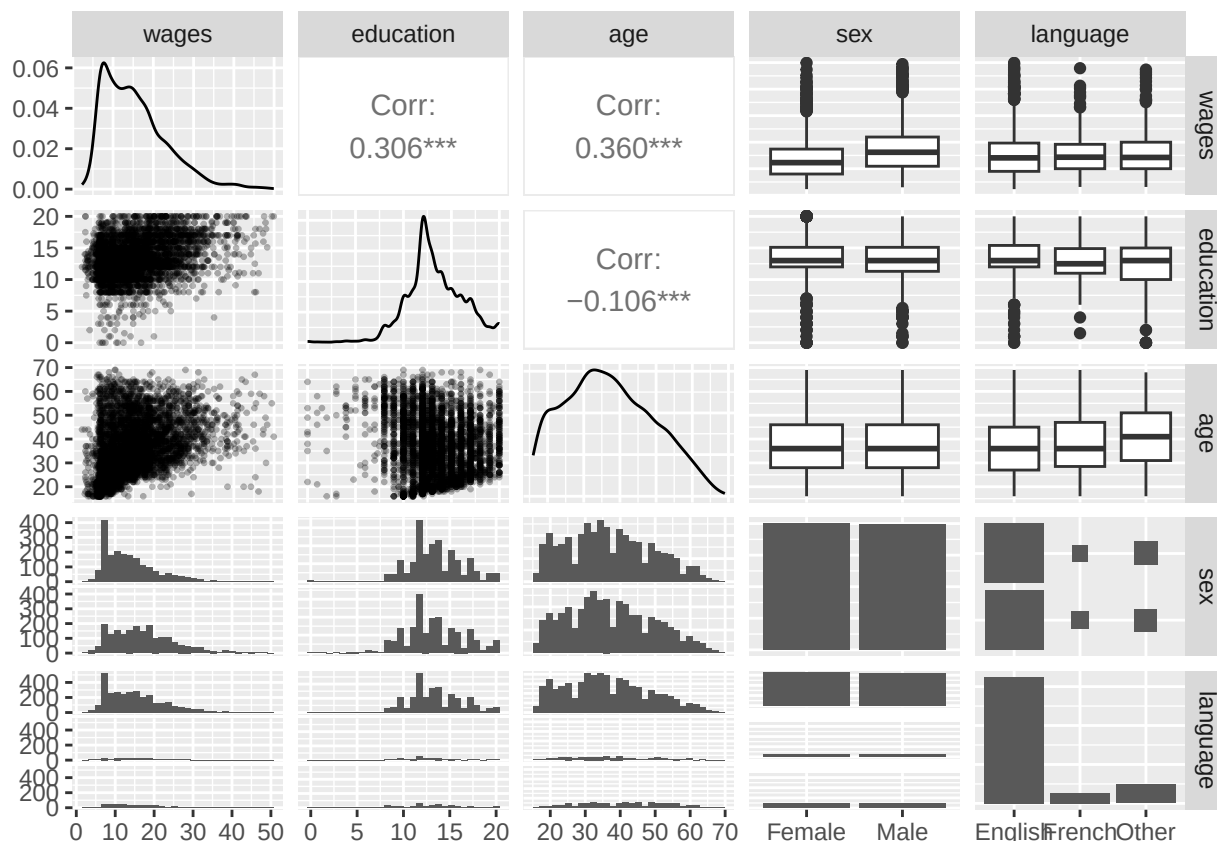
Part 2: Linear regression with the mylm package	4
Statistical Formulas and Theory	4
a) & b) Fitting the model and calculating standard errors	4
c) Residuals plot	5
d) ANOVA and Chi-squared test	6
e) Coefficient of determination (R^2)	6

#Part 1: Explanatory analysis of the dataset First of all we import the dataset and prepare the data for the analysis

```
library(car)
data(SLID, package = "carData")
SLID <- SLID[complete.cases(SLID), ]
```

Afterwards we draw the draw a matrix of diagnostic plots of all the variables

```
ggpairs(SLID, lower = list(continuous = wrap("points", alpha = 0.3, size=0.5)))
```



Looking at the plot, there appear to be some noteworthy relationships between pairs of variables:

- **education** and **wages** have a moderate positive correlation of 0.306 that is, however, highly statistically significant. The same applies to **age** and **wages**.
- median wages are noticeably higher for male individuals compared to females, while they are effectively the same in relation to the language. This suggests that an individual's gender might be strongly related with their wage, while their language has a likely much weaker effect. In both cases, however, we can see that most wages are concentrated towards relatively low values, with the exception of a few, much higher outliers. This is further confirmed by observing the **wages** density plot, which is skewed to the right.
- **age** and **education** have a very weak, but highly statistically significant, negative correlation of -0.106. This means that these two variables are most likely uncorrelated.
- median education is effectively the same for male and female individuals, and very similar for individuals of different languages. This intuitively suggests that education is unaffected by these two variables. In both cases, the median education is relatively high, with the exception of a few outliers with a significantly lower value.
- a similar argument applies to the **age** variable, although it is worth noticing that the median age of individuals of **other** language appears to be higher.
- there does not seem to be anything noteworthy about the relationship between age and language. It can be noticed, however, that the split between male and female individuals seems almost equal, while the **language** variables is dominated by **English**.

To carry out a multiple linear regression analysis of y on the covariates $x_1, \dots, x_p$ we need the following assumptions.

- **Linearity and additivity.** The expected response is a linear function of the regression parameters and the effects of the covariates add up:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, \dots, n,$$

or in matrix form $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $E(\boldsymbol{\varepsilon}) = \mathbf{0}$.

- **Homoscedasticity.** The errors have a constant variance, $\text{Var}(\varepsilon_i) = \sigma^2$ for all i , which does not depend on the covariates.
- **Uncorrelated errors.** $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$, so that $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$.
- **Full rank design.** The design matrix $\mathbf{X}$ has full column rank $p < n$, so that $\mathbf{X}^\top \mathbf{X}$ is invertible and $\hat{\boldsymbol{\beta}}$ is uniquely defined. The covariates are also assumed to be fixed (measured without error).
- **Normality.** For the *normal* linear regression model we additionally assume $\varepsilon_i \sim N(0, \sigma^2)$, independently. This is what makes the exact t - and F -tests valid. It is not needed for $\hat{\boldsymbol{\beta}}$ to be unbiased or for the least squares estimator to be BLUE (Gauss–Markov), and it is not needed for the *asymptotic* z - and χ^2 -tests used in this exercise, which only rely on a large sample size.

Given the strong right-skewness of `wages` noted above, the normality assumption is the one most likely to be violated here — which is a further reason for preferring the asymptotic tests implemented in `mylm`.

Part 2: Linear regression with the mylm package

Statistical Formulas and Theory

The `mym` package implements ordinary least squares (OLS) for a linear model $Y = X\beta + \varepsilon$. The functions in the package are built upon the following formulas:

- **Coefficient estimates:** $\hat{\beta} = (X^\top X)^{-1} X^\top Y$
- **Fitted values:** $\hat{Y} = X\hat{\beta}$
- **Residuals:** $e = Y - \hat{Y}$
- **Residual Sum of Squares (SSE):** $SSE = e^\top e$
- **Degrees of Freedom (df):** $df = n - p$ (where p is the number of parameters).
- **Total Sum of Squares (SST):** $SST = \sum (y_i - \bar{y})^2$
- **Covariance matrix:** $\widehat{\text{Cov}}(\hat{\beta}) = \hat{\sigma}^2 (X^\top X)^{-1}$, with $\hat{\sigma}^2 = \frac{SSE}{n-p}$
- **Standard Errors (SE):** $\text{SE}(\hat{\beta}_j) = \sqrt{\widehat{\text{Cov}}(\hat{\beta})_{jj}}$
- **Asymptotic z-test:** $z_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}$ with p-value $2\Phi(-|z_j|)$.
- **Asymptotic χ^2 -test:** $\chi^2 = \frac{SSE_{\text{reduced}} - SSE_{\text{full}}}{\hat{\sigma}_{\text{full}}^2}$
- **Coefficient of determination:** $R^2 = 1 - \frac{SSE}{SST}$

a) & b) Fitting the model and calculating standard errors

We fit a simple linear regression model to the data with `wages` as the response and `education` as the predictor.

```
modell1 <- mylm(wages ~ education, data = SLID)
modell1b <- lm(wages ~ education, data = SLID)

print.mylm(modell1)
```

```
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
## Coefficients:
##      (Intercept) education
## [1,]      4.971691  0.7923091
```

```
print(modell1b)
```

```
##
## Call:
## lm(formula = wages ~ education, data = SLID)
##
## Coefficients:
## (Intercept)      education
##      4.9717      0.7923
```

The output confirms that our `mym` function yields the exact same coefficient estimates as the standard `lm` function.

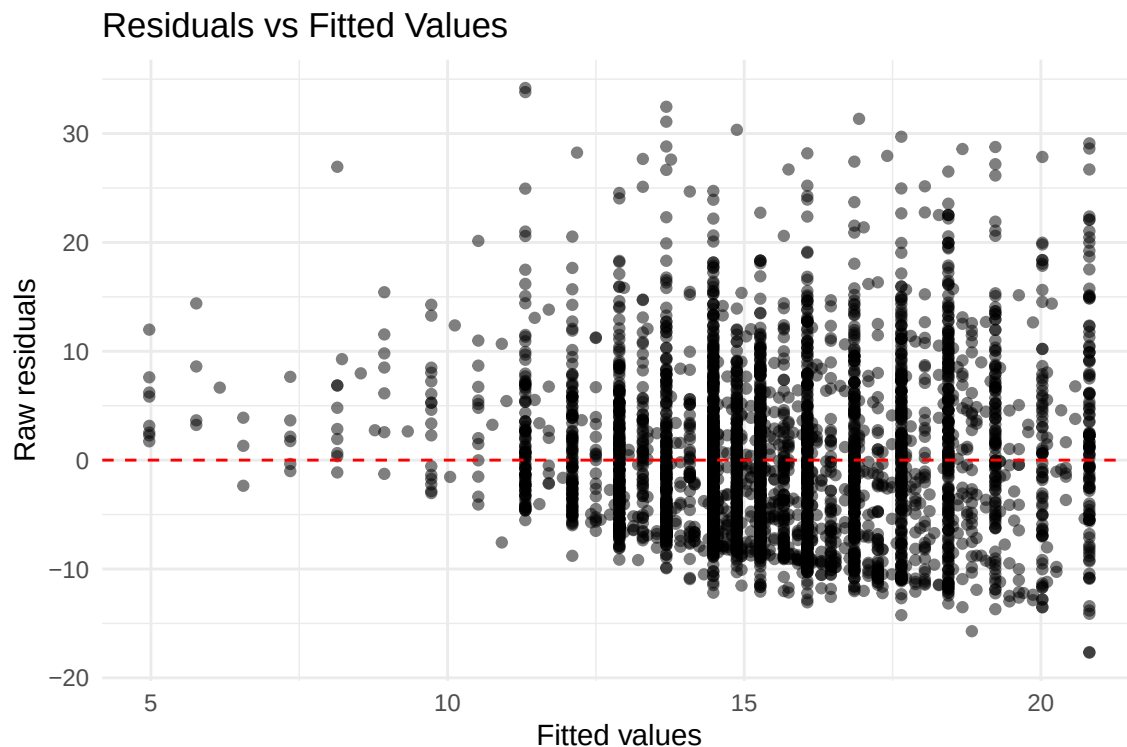
```
summary.mylm(model1)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
## Coefficients:
##              Std. Error
## (Intercept) 4.97169    0.53429  9.3052 < 2.2e-16 ***
## education   0.79231    0.03906 20.2842 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.093586
```

Interpretation of parameters: * **Intercept** ($\hat{\beta}_0 \approx 4.97$): The expected hourly wage for a person with 0 years of education. This is extrapolated outside the data range and holds no practical meaning since wages cannot be negative. * **Education** ($\hat{\beta}_1 \approx 0.79$): For each additional year of education, the expected hourly wage increases by 0.79 units. * **Significance (z-test)**: The z-value for education is 20.28, yielding a p-value strictly close to 0. We confidently reject the null hypothesis ($H_0 : \beta_1 = 0$); education is a highly significant predictor.

c) Residuals plot

```
plot.mylm(model1)
```



Comment on the plot: The residuals versus fitted values plot shows a funnel shape, meaning the variance of the residuals increases as the fitted values increase. This violates the assumption of homoscedasticity.

d) ANOVA and Chi-squared test

```
results_anova <- anova.mylm(model1)
```

```
## Analysis of Variance Table
## Response: wages
##           Df Sum sq X2 value Pr(>X2)
## education 1  23096.20 411.45  1.7747e-91
```

Relationship between tests: In simple linear regression, the χ^2 statistic with 1 degree of freedom is equal to the square of the z -statistic for the slope coefficient. From our summary, $z \approx 20.284$, and $z^2 \approx 411.45$, matching the χ^2 value in the ANOVA table. The critical value for a 5% significance level is $z_{0.025} \approx 1.96$ and $\chi^2_{1,0.05} \approx 3.84$.

e) Coefficient of determination (R^2)

The coefficient of determination R^2 is **0.0936**.

Interpretation: Approximately **9.36%** of the total variability in hourly wages is explained by the variance in years of education. The remaining **90.64%** is unexplained by this model.